Week 5 DSC 180 Presentation



DSD Team
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Problem Statement: Developmental Services Department

- **Stakeholder: Developmental Services Department**
  - The City of San Diego's DSD reviews plans and issues permits for construction and development projects.
  - Application gets submitted → plans are reviewed by respective experts → permit issued

- **Problem: Delays in Permit Approvals**
  - High resubmittal cycles → unclear comments and application requirements
  - Takes more review time for the City and more waiting time for applicant

Field Trip to City Hall (10/12)

- Aligned with David on problem scope and possible solutions
- Approaching problem from applicant's side:
  - Financial motivation to obtain permit ASAP
  - Scattered information for applicants to meet all application requirements in one go
  - Unclear permit approval estimation times
- Established further connection with Mazen to set up meeting and access data

Current System & Preliminary Findings

What the System Is Like Today

- Applicants often don't know what their application requires to get approved.
- Requirements are spread across many sources
- The result is repeated resubmittals, with time and money spent without progress.
- Applicants become frustrated, and the City spends staff time and resources on repeat reviews.

Initial EDA

- Of 168,278 finished record-and-discipline reviews, 53.9% needed at least one resubmittal.
- Only 46% were approved in a single round, and about 5% took five or more rounds.
- Permits requiring plan review take about 100 days on average.
- Frequent filers receive building permits in about 20 days vs ~100 days for everyone else.
- Experience matters, which suggests knowing what to include is the hardest part of the process.

Data - What we have

We now have access to the **DSD Permitting Research Extract:**

- 10 tables with ~2 million rows dated Jan 2023-Oct 2026
- Columns for projects, approvals, reviews, possession history, inspections, and monthly capacity
- Cleanly trackable resubmit cycles with column `DISCIPLINE_CYCLE_NUMBER`

We still need to resolve:

- Comments access
- Compute
- Meeting Tuesday with David and Mazen

Proposed Solution

- A recommendation tool that helps applicants prepare for their specific permit type before they submit, so they know what to expect.
- Flow: past review data → ML model (decision tree) → risk-based tips for the submitter.

Proposed Process

- Train a model on the review data we received. For each record and discipline, the target is the number of review rounds, or simply whether a resubmittal was needed.
- Look for patterns by permit type and location.
- Test on records the model hasn't seen
- Currently leaning toward a decision tree, though we're still exploring.

Proposed Solution

Possible recommendation given current data

- Risk by reviewer → “Similar permits usually have a 70% chance of a Structural resubmittal”
  - Supported by the Reviews data, which records each review’s outcome (Recheck Required or Signed Off) by permit type, location, and discipline.

Out of scope recommendations given current data

- Specific missing items → “X part is missing from this file”
  - Requires reviewer comments, correction reasons, and the list of submitted documents, none of which we currently have.
  - The data only shows that a recheck was needed, not why.

Getting missing data

- Request review comments and correction reasons from the DSD team.
- Web scraping as a final resort (making sure to check with DSD team)

Impact

- Reduces friction on both sides: easier for submitters and applicants, and less time and money spent by both the applicant and the City.
- Faster turnaround, not just for this permit but for others too, because fewer repeat reviews free up reviewers.
- A speedup in the overall permitting system.
- As we saw, frequent filers get building permits in about 20 days versus about 100 for everyone else. If we can get people closer to the 20 day waiting period that would be a significant improvement.

Candidate Deliverables

- **Comments tool**
  - User facing, suggests review points before an applicant actually submits the application for a human to review
  - Dependent on receiving access to comment data
- **Deep Dive Analysis**
  - Fully fleshed out report that reviews where the existing deficiencies might be and if optimizing resource allocation could improve speed
  - Possible with current data extract
- **Consolidated Reference Database**
  - One unified place where any applicant can see all of the information they need with minimal searching

Target Metrics

- Application completeness & quality
  - Increase percentage of submissions accepted after the pre-screen and fee payment
- Lower resubmittal cycle
  - Identify projects with most resubmittal cycles and track to see if our solution helps reduce common mistakes that cause resubmittal
  - Percentage of resubmittal cycle (reduce % of 2+, 5+, etc)

DSD Meeting

Meeting with DSD on Tuesday 10/27 to review:

- Comments data access
- Vetting proposed deliverables
- Finalize specifications for model prototypes

Timeline

- **Week 4: 10/22 (Thu)** : Received the data.
- **Week 5: 10/25 (Sun)** : Cleaned the Reviews data and built the review-cycle table (done).
- **Week 5: 10/26 (Mon)** : Check in with Ben (TA) and Bijan / Capstone Panel.
- **Week 5: 10/27 (Tue)** : Meet with the DSD team, David and Mazen. Ask whether review comments and correction reasons can be shared.
- **Week 5: 10/29 (Thu)** : Preliminary model ready to test.